A Gumbel law for Markovian super-coupon collection

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Abstract

An r-element subset of an n-element universe evolves by replacing a uniformly chosen element with a uniformly chosen element outside the current set. Each configuration collects all of its s-element subsets. For every fixed pair $2 \le s < r$, we prove that the expected time to collect all s-subsets is asymptotic to

$$\frac{\binom{n}{s}}{\binom{r-1}{s-1}} \log \binom{n}{s}.$$

This proves the fixed-parameter off-diagonal case of the conjecture of Athreya, Mukherjee and Mukherjee. We also obtain a Poisson limit for the number of subsets still missing near completion and a standard Gumbel limit for the centered and scaled collection time. The conclusions hold from every initial configuration. The proof controls joint avoidance of any fixed family of target subsets, uniformly over their overlaps. Restricting the walk to the labels in that family gives a finite transition matrix, and a stationary-flow identity determines its leading killing rate. Standard spectral and factorial-moment arguments then give the collection-time limits.

Keywords. Coupon collector; Johnson graph; cover time; Gumbel distribution; hitting time; finite Markov chain.

1 Introduction and main result

Suppose a collection contains r distinct labels and changes by one random replacement at each step. A configuration reveals all of its s-element subsets, and the collection is complete when every s-subset of the underlying universe has appeared. Successive configurations share $r-1$ labels, so many of the subsets observed in one step are observed again in the next. The question is how this dependence changes the time needed to see the last missing subset.

Athreya, Mukherjee and Mukherjee introduced this Markovian super-coupon model and conjectured its leading expected collection time [1, Conjecture 2.4]. They proved the expectation asymptotic in the diagonal case $s = r$, where collection is vertex coverage of a Johnson graph [1, Theorem 2.5]. We prove the conjectured formula for every fixed $2 \le s < r$ and determine the limiting fluctuations. In this regime a configuration collects several overlapping subsets at once; the theorem below gives a Gumbel limit for completion and a Poisson limit for the number still missing. We use the corrected version of [1]; see also [2].

Fix integers $2 \le s < r$. For $n > r$, write $[n] = \{1, \dots, n\}$ and let $(R_t)_{t \ge 0}$ be the following discrete-time walk on $\binom{[n]}{r}$. Given R_t, choose a uniformly from R_t and b uniformly from $[n] \setminus R_t$, independently, and set

$$R_{t+1} = (R_t \setminus \{a\}) \cup \{b\}.$$

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The insertion is from the complement of the old configuration; in particular, the removed element cannot be reinserted in that step. Each edge of the Johnson graph has transition probability $1/[r(n-r)]$, so the uniform law π on $\binom{[n]}{r}$ is stationary. Define

$$U_t = \#\{S \in \tbinom{[n]}{s} : S \not\subseteq R_u \text{ for every } 0 \le u \le t\}, \tag{1.1}$$

$$T_n = \inf\{t \ge 0 : U_t = 0\}. \tag{1.2}$$

Thus the initial configuration already contributes its subsets. All logarithms are natural, and all limits below are as $n \to \infty$ with r, s fixed. Put

$$N_n = \binom{n}{s}, \qquad b_{r,s} = \binom{r-1}{s-1}, \qquad \lambda_n = \frac{b_{r,s}}{N_n}. \tag{1.3}$$

Theorem 1.1. *For every fixed $2 \le s < r$ and every initial distribution of R_0, the following assertions hold. For each fixed $x \in \mathbb{R}$, let*

$$t_n(x) = \left\lfloor \lambda_n^{-1}(\log N_n + x) \right\rfloor.$$

Then

$$U_{t_n(x)} \xrightarrow{d} \text{Poisson}(e^{-x}), \qquad \mathbb{P}\{\lambda_n T_n - \log N_n \le x\} \longrightarrow e^{-e^{-x}}. \tag{1.4}$$

Moreover,

$$\frac{T_n}{\lambda_n^{-1} \log N_n} \xrightarrow{\mathbb{P}} 1, \qquad \mathbb{E}T_n \sim \lambda_n^{-1} \log N_n = \frac{r}{s}\binom{r}{s}^{-1}\binom{n}{s}\log\binom{n}{s}. \tag{1.5}$$

The last identity follows from $\binom{r-1}{s-1} = (s/r)\binom{r}{s}$ and gives exactly the coefficient in [1, Conjecture 2.4]. Their convention counts the initial draw as round one, adding one to T_n; this does not affect the limits. For example, when $s = 2$,

$$\mathbb{E}T_n \sim \frac{n^2 \log n}{r-1}. \tag{1.6}$$

The result therefore determines both the leading time to see every pair in a changing r-set and its fluctuations on the smaller scale $\binom{n}{2}/(r-1)$.

The coefficient has a local interpretation already given in [1, Remark 2.2]. A swap creates exactly $b_{r,s}$ subsets that were absent from the preceding configuration. These subsets may have appeared earlier in the trajectory, so that count alone does not determine the completion time. For a fixed target S, stationarity gives

$$\mathbb{P}_\pi(S \subseteq R_t) = \frac{\binom{r}{s}}{\binom{n}{s}}, \qquad \mathbb{P}_\pi(S \subseteq R_t,\ S \not\subseteq R_{t+1}) = \frac{s}{r}\frac{\binom{r}{s}}{\binom{n}{s}} = \lambda_n. \tag{1.7}$$

The second equality holds because the target is lost precisely when one of its s labels is removed. The central estimate below shows that the corresponding rate for avoiding any fixed k distinct targets is $k\lambda_n + O(n^{-s-1})$, even when the targets overlap. This joint estimate controls the last few missing subsets.

Relation to hitting-time methods

Exponential approximation for hitting times, target contraction and killed transition matrices are established tools. Manzo, Quattropani and Scoppola [7] give a probabilistic proof of the First Visit Time Lemma and explicitly treat contraction of target sets in their Section 2.3. Related return-time calculations appear in the hypergraph covering and inform-time analysis

of Cooper, Frieze and Radzik [5]. The classical multiplicative comparison of Aldous and Brown is presented, together with a proof and a discrete-time discussion, by Berestycki, Hermon and Teyssier [4, Theorem 1.1 and Remark 2.6]. Spectral perturbation also appears in probabilistic marking problems for graph walks [3].

Our contribution is the collection-time theorem for the dependent subset model. Its proof supplies a uniform killing-rate estimate for overlapping target families. General hitting-time theory can also be applied once the model's short-return probabilities have been estimated; we give a self-contained finite-matrix argument to make the required precision explicit. The original paper's diagonal proof already uses an overlap-count chain. Here we retain the individual labels in a fixed family of targets, which allows the same avoidance calculation to include every overlap pattern.

Section 2 proves the joint avoidance estimate. Section 3 applies it to the missing-subset count and then controls the expectation tail. Constants may depend on the fixed integers r, s and, for a family of k targets, on k; they do not depend on n or on the target labels.

2 Joint avoidance of target subsets

We first explain why stationarity can be used without restricting the theorem. If R and R' are two initial r-sets, a permutation of $[n]$ taking R to R' preserves the transition rule and permutes the complete family of s-subsets. It therefore preserves the entire process $(U_t)_{t\geq 0}$ in law. Every deterministic initial configuration gives the same law for this process, and every initial distribution is a mixture of those laws. It is consequently enough to prove Theorem 1.1 under π. This symmetry does not hold for the avoidance of a specified family of named targets, for which stationarity will remain an explicit hypothesis.

Fix $k \geq 1$ and distinct $S_1, \dots, S_k \in \binom{[n]}{s}$. Write $\mathcal{S} = (S_1, \dots, S_k)$ and define

$$p_{n,\mathcal{S}}(t) = \mathbb{P}_\pi\{S_j \not\subseteq R_u \text{ for all } 1 \leq j \leq k,\ 0 \leq u \leq t\}.$$

Proposition 2.1. *For fixed r, s, k, there are constants $C < \infty$, $q \in (0,1)$ and n_0 such that, for every $n \geq n_0$ and every such family $\mathcal{S}$, there are numbers $a_{n,\mathcal{S}}$ and $\rho_{n,\mathcal{S}} \in (0,1)$ satisfying*

$$|a_{n,\mathcal{S}} - 1| \leq Cn^{-1}, \tag{2.1}$$

$$1 - \rho_{n,\mathcal{S}} = k\lambda_n + O(n^{-s-1}), \tag{2.2}$$

$$\left|p_{n,\mathcal{S}}(t) - a_{n,\mathcal{S}}\rho_{n,\mathcal{S}}^t\right| \leq Cq^t \qquad (t \in \mathbb{Z}_{\geq 0}). \tag{2.3}$$

The error in (2.2) *is uniform over the target family.*

The exponential form by itself is insufficient at time $\lambda_n^{-1} \log N_n$: an error in the rate is multiplied by that time. The relative error $O(n^{-1})$ in (2.2) contributes only $O((\log n)/n)$ to the exponent. We prove the proposition by separating this rate calculation from the finite-dimensional spectral estimate.

2.1 The marked-subset chain

Let $V = \bigcup_{j=1}^k S_j$ and $m = |V| \leq ks$. A label in V will be called marked. The process

$$A_t = R_t \cap V$$

records all information needed to decide whether a target has appeared. For $n \geq m + r$, its state space is the fixed finite set

$$\Omega = \{A \subseteq V : |A| \leq r\}.$$

Given $A \in \Omega$ with $d = |A|$, the configuration contains $r - d$ unmarked labels and its complement contains $n - m - r + d$ unmarked labels. Counting the possible removals and insertions gives the quotient transition matrix P_n:

$$\begin{aligned} P_n(A, A \cup \{b\}) &= \frac{r-d}{r(n-r)}, && b \in V \setminus A, \\ P_n(A, A \setminus \{a\}) &= \frac{n-m-r+d}{r(n-r)}, && a \in A, \\ P_n(A, (A \setminus \{a\}) \cup \{b\}) &= \frac{1}{r(n-r)}, && a \in A,\ b \in V \setminus A, \\ P_n(A, A) &= \frac{(r-d)(n-m-r+d)}{r(n-r)}. \end{aligned} \tag{2.4}$$

The first line is omitted when $d = r$. These probabilities depend only on A, which proves that (A_t) is a Markov chain with the stated kernel. Its stationary distribution, induced by π, is

$$\nu_n(A) = \frac{\binom{n-m}{r-|A|}}{\binom{n}{r}}. \tag{2.5}$$

Let

$$H = \{A \in \Omega : S_j \subseteq A \text{ for some } j\}, \qquad D = \Omega \setminus H,$$

and let $Q_n = (P_n)_{DD}$ be the matrix restricted to the surviving states. Its rows may sum to less than one because transitions into H have been removed. If $\nu_{n,D}$ denotes the row vector of stationary masses on D, then

$$p_{n,\mathcal{S}}(t) = \nu_{n,D} Q_n^t \mathbf{1}. \tag{2.6}$$

Restricting the initial vector to D accounts for observation at time zero; the power t accounts for the next t transitions.

As $n \to \infty$, insertion of a marked label has probability tending to zero. The limiting matrix on D is

$$Q_\infty(A, A) = 1 - |A|/r, \qquad Q_\infty(A, A \setminus \{a\}) = 1/r \quad (a \in A). \tag{2.7}$$

The set D is closed under deletion. Thus Q_∞ is stochastic, and its unique absorbing state is $\varnothing$. Ordering states by cardinality makes it triangular, with one diagonal entry equal to one and all other diagonal entries among $1 - d/r$, $1 \le d \le r$. In particular, the eigenvalue one is simple, and its spectral projection is

$$\Pi_\infty = \mathbf{1} e_\varnothing^{\mathsf{T}}, \tag{2.8}$$

where $e_\varnothing$ is the coordinate vector of the empty state.

2.2 Spectral separation

For completeness, we give the finite-dimensional perturbation argument used here. It is the usual contour construction for spectral projections; see Kato [6, Chapter II] for the general framework. For a fixed target pattern, the entries in (2.4) are analytic functions of $z = 1/n$ near zero. Choose a small circle Γ around one that excludes the other eigenvalues of Q_∞. For all sufficiently small z, the resolvent exists on Γ, and

$$\Pi_n = \frac{1}{2\pi i} \int_\Gamma (wI - Q_n)^{-1} \, \mathrm{d}w = \Pi_\infty + O(n^{-1})$$

is a rank-one projection. It corresponds to a simple real eigenvalue $\rho_n = 1 + O(n^{-1})$; reality follows because Q_n is real and there is only one eigenvalue inside Γ. Applying Π_n to $\mathbf{1}$ and normalizing its empty-state coordinate gives a right eigenvector

$$Q_n f_n = \rho_n f_n, \qquad f_n(\varnothing) = 1, \qquad f_n = \mathbf{1} + O(n^{-1}). \tag{2.9}$$

In particular, f_n is positive for large n.

Choose a fixed q with $1 - 1/r < q < 1$. For large n, all eigenvalues other than ρ_n lie strictly inside $|w| = q$, and $\rho_n > q$. The resolvents on this circle are uniformly bounded, so

$$Q_n^t = \rho_n^t \Pi_n + E_{n,t}, \qquad E_{n,t} = \frac{1}{2\pi i} \int_{|w|=q} w^t (wI - Q_n)^{-1} \, \mathrm{d}w, \qquad \|E_{n,t}\| \le C q^t. \tag{2.10}$$

The bound does not require diagonalizability of the remaining spectrum. From (2.5), $\nu_{n,D} = e_\varnothing^{\mathsf{T}} + O(n^{-1})$. Combining (2.6) and (2.10) proves (2.3) with

$$a_{n,\mathcal{S}} = \nu_{n,D} \Pi_n \mathbf{1} = 1 + O(n^{-1}).$$

There are only finitely many families of k distinct s-sets up to relabeling of at most ks marked elements. Taking the largest constants and thresholds over these patterns makes every estimate uniform in $\mathcal{S}$. It remains to determine $1 - \rho_n$ to the accuracy in (2.2).

2.3 The killing rate from stationary flow

Write P_{HD} for the transitions of P_n from H to D. Stationarity, restricted to columns in D, gives

$$\nu_{n,D} Q_n + \nu_{n,H} P_{HD} = \nu_{n,D}.$$

Multiplication by f_n turns this balance of incoming and outgoing mass into the exact identity

$$(1 - \rho_n) \nu_{n,D} f_n = \nu_{n,H} P_{HD} f_n. \tag{2.11}$$

Let $F_n = \nu_{n,H} P_{HD} \mathbf{1}$ be the stationary probability of leaving the union of target events in one step. Since $\nu_n(H) = O(n^{-s})$ and (2.9) holds uniformly on D, we have $\nu_{n,D} f_n = 1 + O(n^{-1})$. Positivity of the transition probabilities also gives $\nu_{n,H} P_{HD} f_n = F_n(1 + O(n^{-1}))$. Dividing in (2.11) yields

$$1 - \rho_n = F_n(1 + O(n^{-1})). \tag{2.12}$$

Thus it suffices to add the individual target exit probabilities and bound the error caused by overlapping or successive target visits.

On the original configuration space, put $H_j = \{R : S_j \subseteq R\}$. Let Z count the individual events H_j exited during a stationary step, and let Y indicate that their union is exited. Equation (1.7) gives $\mathbb{E}_\pi Z = k\lambda_n$, whereas $\mathbb{E}_\pi Y = F_n$. Also $Z \ge Y$. If at most one target is present initially, a difference between Z and Y can occur only by leaving that target and entering a different one. Consequently,

$$\begin{aligned} 0 \le k\lambda_n - F_n \le{}& k \mathbb{P}_\pi\{R_t \text{ contains at least two targets}\} \\ &+ \sum_{i \ne j} \mathbb{P}_\pi\{R_t \in H_i \setminus H_j,\ R_{t+1} \in H_j\}. \end{aligned} \tag{2.13}$$

For distinct targets, $|S_i \cup S_j| \ge s + 1$. If this union has size $\ell \le r$, then

$$\mathbb{P}_\pi\{S_i \cup S_j \subseteq R_t\} = \frac{\binom{r}{\ell}}{\binom{n}{\ell}} = O(n^{-s-1});$$

the probability is zero when $\ell > r$. A union bound over pairs controls the first term in (2.13). For the second term, a target S_j absent from R_t can be formed in one step only if it has exactly one missing label, which must be the inserted element. Conditional on any such R_t, this has probability at most $1/(n-r)$. Therefore each summand is bounded by

$$\frac{\pi(H_i)}{n-r} = \frac{1}{n-r}\frac{\binom{r}{s}}{\binom{n}{s}} = O(n^{-s-1}).$$

Since k is fixed, (2.13) yields

$$F_n = k\lambda_n + O(n^{-s-1}). \tag{2.14}$$

Combining (2.12) and (2.14), and using $\lambda_n = \Theta(n^{-s})$, proves (2.2). It also shows $\rho_n \in (0,1)$ for all sufficiently large n. This completes the proof of Proposition 2.1.

3 From avoidance to completion

The joint estimate can now be summed over target families. Fix $x \in \mathbb{R}$ and $k \geq 1$. At $t = t_n(x)$, equation (2.2) and the expansion of the logarithm give, uniformly over $\mathcal{S}$,

$$t \log \rho_{n,\mathcal{S}} = -k(\log N_n + x) + O\Big(\frac{\log n}{n}\Big) + O(n^{-s}\log n). \tag{3.1}$$

Integer rounding contributes only $O(\lambda_n)$, which is absorbed in the displayed error. The term q^t in (2.3) is smaller than every fixed power of n^{-1} at these times. Together with (2.1), this proves

$$p_{n,\mathcal{S}}(t_n(x)) = (1+o(1))N_n^{-k}e^{-kx}, \tag{3.2}$$

with relative error uniform over all ordered distinct k-tuples of targets.

For a nonnegative integer u, write $(u)_k = u(u-1)\cdots(u-k+1)$. By (1.1),

$$\mathbb{E}_\pi(U_{t_n(x)})_k = \sum_{\substack{(S_1,\dots,S_k)\\ \text{ordered and distinct}}} p_{n,\mathcal{S}}(t_n(x)) \longrightarrow e^{-kx}. \tag{3.3}$$

There are $(N_n)_k$ terms in this sum. They need not be equal, because the overlap patterns differ; their common asymptotic value follows from the uniformity in (3.2). These are the factorial moments of Poisson(e^{-x}). The factorial-moment criterion gives the first limit in (1.4): ordinary moments are finite linear combinations of factorial moments, and the Poisson law is moment-determinate. Since U_t is integer-valued,

$$\mathbb{P}_\pi(T_n \leq t_n(x)) = \mathbb{P}_\pi(U_{t_n(x)} = 0) \longrightarrow e^{-e^{-x}},$$

which proves the Gumbel limit. Tightness of $\lambda_n T_n - \log N_n$ then gives the convergence in probability in (1.5), because $\log N_n \to \infty$.

To prove the expectation asymptotic, we use the all-time estimate rather than infer convergence of means from weak convergence. For a single target, permutation symmetry makes the principal eigenvalue independent of its labels; denote it by $\rho_n^{(1)}$. Proposition 2.1 and a union bound give

$$\mathbb{P}_\pi(T_n > t) \leq CN_n((\rho_n^{(1)})^t + q^t), \qquad 1-\rho_n^{(1)} = \lambda_n(1+O(n^{-1})). \tag{3.4}$$

Fix $\varepsilon > 0$ and put $t_+ = \lceil (1+\varepsilon)\lambda_n^{-1}\log N_n \rceil$. Summing the two geometric tails in (3.4) yields

$$\begin{aligned}\sum_{t \geq t_+} \mathbb{P}_\pi(T_n > t) &\leq \frac{CN_n(\rho_n^{(1)})^{t_+}}{1-\rho_n^{(1)}} + \frac{CN_n q^{t_+}}{1-q} \\ &= O\Big(\lambda_n^{-1}N_n^{-\varepsilon+o(1)}\Big) + o(1) = o(\lambda_n^{-1}\log N_n).\end{aligned} \tag{3.5}$$

Hence $\mathbb{E}_\pi T_n \le (1+\varepsilon+o(1))\lambda_n^{-1}\log N_n$. For $t_- = \lfloor (1-\varepsilon)\lambda_n^{-1}\log N_n \rfloor$ with $0<\varepsilon<1$, convergence in probability gives

$$\mathbb{E}_\pi T_n \ge t_- \mathbb{P}_\pi(T_n > t_-) = (1-\varepsilon+o(1))\lambda_n^{-1}\log N_n.$$

Letting $\varepsilon \downarrow 0$ proves the mean asymptotic. The symmetry argument at the start of Section 2 transfers all these conclusions from π to every initial distribution and completes the proof of Theorem 1.1.

The uniformity used here concerns any fixed number of targets and fixed r, s. The dimension of the marked-subset chain and the perturbation constants depend on those parameters, so the proof makes no assertion when r or s grows with n.

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